Phase 4 — research.py + LLM-Kandidaten

DAX-CFD Intraday-Bot · erste AI-Phase · Research-Orchestrator Research.run()

live-verified · 2026-06-09

88 mocked tests · live_test 3/3

CODE (deterministisch) · AI (einziger LLM-Call) · Input / Guard · kein Trade

P1 broker_wrapper
get_price · get_market_info · get_ohlcv · oper

P2 persistence
trades · lessons · score · save_candidates

P3 external_data
get_brain_context (~15 min delayed)

Dependency Injection (Konstruktor) — Phase 4 importiert keine Schwester-Packag

1 · Preflight (Session-Health)
is_connected/connect · get_account.ok · Anker get_price TRADEABLE

2 · build_context (P1 probe · P2 read · P3 brain)
epic_allowlist proben · is_tradeable · None-tolerant

3 · threshold + resolve_allowed_directions
Floor 55 / 70 (score<50) · Long-Bias-Clamp → ("BUY",) bei score<50

4 · build_prompt → (system, user, json_schema)
epic-Enum = reale Liste (anyOf-nullable) · Delay ehrlich markiert

★ 5 · llm.ask_candidate() — AI
Messages API · Structured Outputs · der EINZIGE LLM-Call im Bot
1 Retry nur bei transient · token_meter (USD+EUR) · kein Caching

AI-Grenze:
nur diese Box

★ 6 · validate_candidate — Halluzinations-Guard
epic∈Universum · direction∈allowed · Floor · Live-Spread-Recheck

7 · apply_filter — DAS reale Gate
Spread · TRADEABLE · direction · Drift-Kohärenz = soft warn (kein Reject)

8 · save_candidates([candidate]) (PASS)
genau 1× save pro Zyklus · stdout = JSON, logs → stderr

no-trade sink
Preflight-Fail
leeres Universum
LLM-Error
REJECT / abstain
Filter-Fail

→ save_candidates([]) · kein Trade

Candidate-Contract (frozen · Phase 4 → 5)

```
epic : str (∈ allowlist)
direction : "BUY" | "SELL"
llm_confidence : 0–100 (advisory!)
reasoning : str
spread_pct_at_pick · drift_at_pick|None
score_at_pick · threshold_applied
generated_at(ISO) · source="research"
```

abstain/reject → leere Liste, kein Feld

Output → data/state/turbo_candidates.json
(Legacy-Name · TTL 30 min · self-clearing)

PASS → [1 Candidate]

sonst → [ ] kein Trade

→ Phase 5 · Gate 2 konsumiert diese Datei
live-tested ≠ profit-validated — Edge unbekannt (Phase 7)

Module: models · validator(18) · context_builder(13) · prompt(10) · llm_client(10) · token_meter(6) · usage_report(4) · candidate_filter(15) · research(